Group 21 Project

Test loading the data

The loader code already has the capability to remove not useful columns, but we need to discuss which ones to remove. For now, we will load all the data and then decide which columns to keep or remove.

```r
source("load.r")
# Load the data frames
# TODO remove not useful columns, it's already in the code but need to discuss which ones to remove
order_items_frame <- load_data("archive/olist_order_items_dataset.csv")
order_seller_frame <- load_data("archive/olist_sellers_dataset.csv")
order_product_frame <- load_data("archive/olist_products_dataset.csv")
order_translation_frame <- load_data("archive/product_category_name_translation.csv")
#display the data for testing
str(order_items_frame)
```

```
## 'data.frame':    112650 obs. of  7 variables:
##  $ order_id           : chr  "00010242fe8c5a6d1ba2dd792cb16214" "00018f77f2f0320c557190d7a144bdd3" "0
##  $ order_item_id      : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ product_id         : chr  "4244733e06e7ecb4970a6e2683c13e61" "e5f2d52b802189ee658865ca93d83a8f" "
##  $ seller_id          : chr  "48436dade18ac8b2bce089ec2a041202" "dd7ddc04e1b6c2c614352b383efe2d36" "5
##  $ shipping_limit_date: chr  "2017-09-19 09:45:35" "2017-05-03 11:05:13" "2018-01-18 14:48:30" "2018-
##  $ price              : num  58.9 239.9 199 13 199.9 ...
##  $ freight_value      : num  13.3 19.9 17.9 12.8 18.1 ...
```

```r
str(order_seller_frame)
```

```
## 'data.frame':    3095 obs. of  4 variables:
##  $ seller_id             : chr  "3442f8959a84dea7ee197c632cb2df15" "d1b65fc7debc3361ea86b5f14c68d2e2
##  $ seller_zip_code_prefix: int  13023 13844 20031 4195 12914 20920 55325 16304 1529 80310 ...
##  $ seller_city           : chr  "campinas" "mogi guacu" "rio de janeiro" "sao paulo" ...
##  $ seller_state          : chr  "SP" "SP" "RJ" "SP" ...
```

```r
str(order_product_frame)
```

```
## 'data.frame':    32340 obs. of  9 variables:
##  $ product_id                : chr  "1e9e8ef04dbcff4541ed26657ea517e5" "3aa071139cb16b67ca9e5dea641a
##  $ product_category_name     : chr  "perfumaria" "artes" "esporte_lazer" "bebes" ...
##  $ product_name_lenght       : int  40 44 46 27 37 60 56 56 57 36 ...
##  $ product_description_lenght: int  287 276 250 261 402 745 1272 184 163 1156 ...
##  $ product_photos_qty        : int  1 1 1 1 4 1 4 2 1 1 ...
##  $ product_weight_g          : int  225 1000 154 371 625 200 18350 900 400 600 ...
##  $ product_length_cm         : int  16 30 18 26 20 38 70 40 27 17 ...
##  $ product_height_cm         : int  10 18 9 4 17 5 24 8 13 10 ...
##  $ product_width_cm          : int  14 20 15 26 13 11 44 40 17 12 ...
##  - attr(*, "na.action")= 'omit' Named int [1:611] 106 129 146 155 198 245 295 300 348 429 ...
##   ..- attr(*, "names")= chr [1:611] "106" "129" "146" "155" ...
```

Create merged data frame

This merges the three data frames (order_items_frame, order_seller_frame, order_product_frame) based on common columns (seller_id and product_id). The resulting merged data frame will contain all the relevant information from the three original data frames, allowing for more comprehensive analysis.

```r
# Merge the data frames based on common columns
merged_frame <- merge_frames(order_items_frame, order_seller_frame, by_column = "seller_id")
merged_frame <- merge_frames(merged_frame, order_product_frame, by_column = "product_id")

# Map Portuguese category names to English using a left join
merged_frame <- merge(
  merged_frame,
  order_translation_frame,
  by = "product_category_name",
  all.x = TRUE
)

# Keep only the English category name in the merged frame
merged_frame$product_category_name <- merged_frame$product_category_name_english
merged_frame$product_category_name_english <- NULL

# Display the structure of the merged data frame
str(merged_frame)
```

```
## 'data.frame':    111046 obs. of  18 variables:
##  $ product_category_name     : chr  "agro_industry_and_commerce" "agro_industry_and_commerce" "agro_i
##  $ product_id                : chr  "672e757f331900b9deea127a2a7b79fd" "672e757f331900b9deea127a2a7b7
##  $ seller_id                 : chr  "e59aa562b9f8076dd550fcddf0e73491" "e59aa562b9f8076dd550fcddf0e73
##  $ order_id                  : chr  "f34349c1787f5a47ace7555af961a47b" "28e69219369607a9fc2403113041
##  $ order_item_id             : int  1 2 2 1 2 1 1 1 6 1 ...
##  $ shipping_limit_date       : chr  "2018-01-15 07:26:26" "2017-09-28 14:30:11" "2018-02-13 20:10:42
##  $ price                     : num  410 398 25 109.9 27.5 ...
##  $ freight_value             : num  55.2 25.7 30.1 15.1 15.3 ...
##  $ seller_zip_code_prefix    : int  82810 82810 82810 13473 82810 82810 5528 13202 90230 13382 ...
##  $ seller_city               : chr  "curitiba" "curitiba" "curitiba" "americana" ...
##  $ seller_state              : chr  "PR" "PR" "PR" "SP" ...
##  $ product_name_lenght       : int  39 39 56 59 48 48 60 50 53 52 ...
##  $ product_description_lenght: int  383 383 93 693 157 157 1843 936 1461 934 ...
##  $ product_photos_qty        : int  1 1 1 6 1 1 3 6 1 3 ...
##  $ product_weight_g          : int  2650 2650 100 1900 250 250 550 6950 1500 50 ...
##  $ product_length_cm         : int  30 30 16 45 17 17 16 36 22 16 ...
##  $ product_height_cm         : int  30 30 5 15 3 3 16 30 16 6 ...
##  $ product_width_cm          : int  20 20 11 40 10 10 16 33 24 13 ...
```

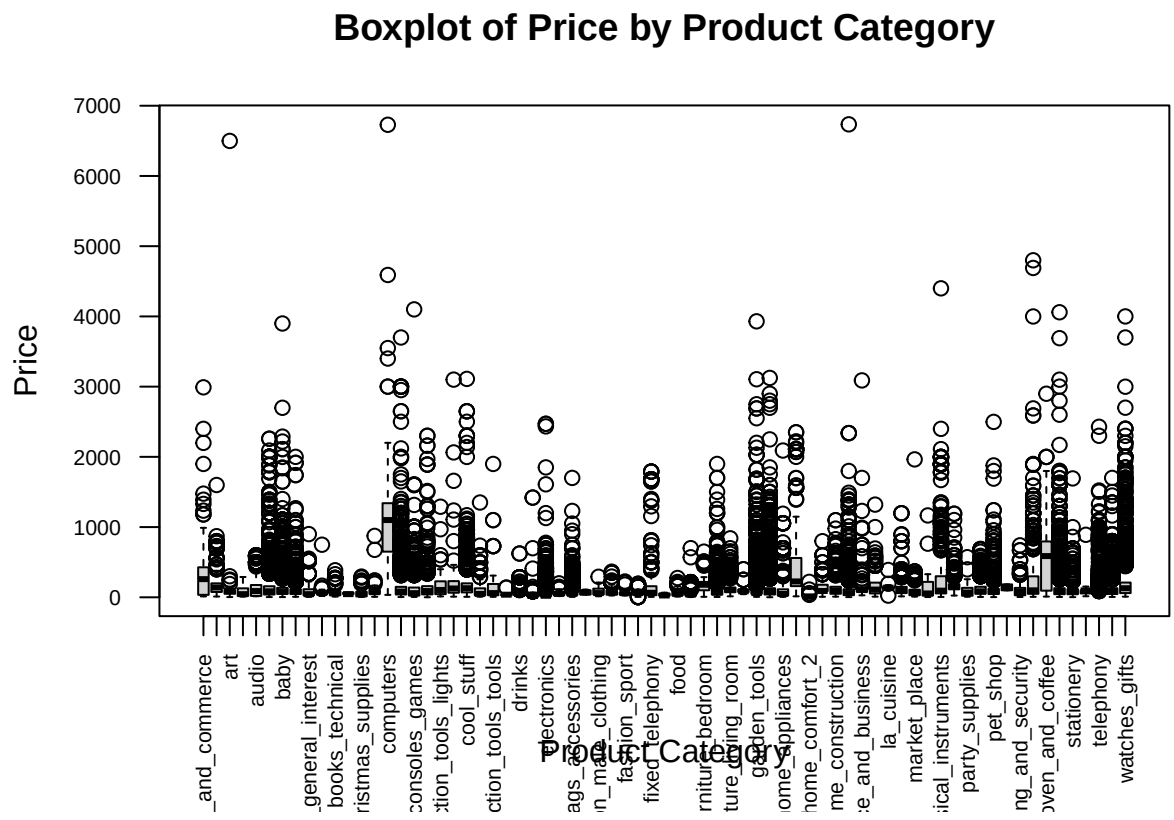
Exploration

Now that we have the merged data frame, we can start exploring the data to gain insights and identify patterns. We can perform various analyses such as summary statistics, visualizations, and correlation analysis to understand the relationships between different variables in the dataset.

```r
# Summary statistics of the merged data frame
summary(merged_frame)
```

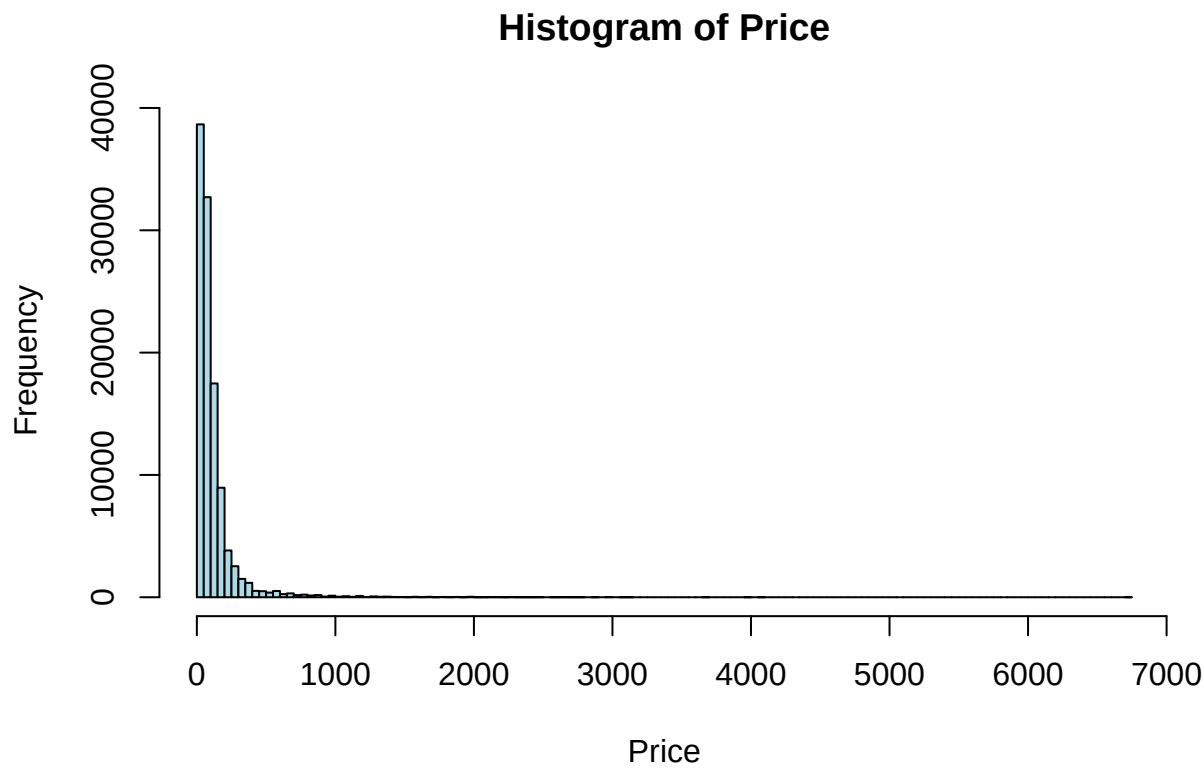
```
##  product_category_name  product_id         seller_id           order_id
##  Length:111046         Length:111046      Length:111046      Length:111046
##  Class :character      Class :character   Class :character   Class :character
##  Mode  :character      Mode  :character   Mode  :character   Mode  :character
##
##
##
##  order_item_id    shipping_limit_date     price         freight_value
##  Min.   : 1.000   Length:111046       Min.   :   0.85   Min.   :  0.00
##  1st Qu.: 1.000   Class :character    1st Qu.:  39.90   1st Qu.: 13.08
##  Median : 1.000   Mode  :character    Median :  74.99   Median : 16.30
##  Mean   : 1.198                       Mean   : 120.76   Mean   : 20.03
##  3rd Qu.: 1.000                       3rd Qu.: 134.90   3rd Qu.: 21.18
##  Max.   :21.000                       Max.   :6735.00   Max.   :409.68
##  seller_zip_code_prefix seller_city        seller_state
##  Min.   : 1001          Length:111046      Length:111046
##  1st Qu.: 6429          Class :character   Class :character
##  Median :13660          Mode  :character   Mode  :character
##  Mean   :24497
##  3rd Qu.:28470
##  Max.   :99730
##  product_name_lenght product_description_lenght product_photos_qty
##  Min.   : 5.00       Min.   :   4.0             Min.   : 1.00
##  1st Qu.:42.00       1st Qu.: 348.0             1st Qu.: 1.00
##  Median :52.00       Median : 603.0             Median : 1.00
##  Mean   :48.78       Mean   : 787.9             Mean   : 2.21
##  3rd Qu.:57.00       3rd Qu.: 987.0             3rd Qu.: 3.00
##  Max.   :76.00       Max.   :3992.0             Max.   :20.00
##  product_weight_g product_length_cm product_height_cm product_width_cm
##  Min.   :    0    Min.   :  7.00    Min.   :  2.00    Min.   :  6.00
##  1st Qu.:  300    1st Qu.: 18.00    1st Qu.:  8.00    1st Qu.: 15.00
##  Median :  700    Median : 25.00    Median : 13.00    Median : 20.00
##  Mean   : 2100    Mean   : 30.21    Mean   : 16.63    Mean   : 23.04
##  3rd Qu.: 1800    3rd Qu.: 38.00    3rd Qu.: 20.00    3rd Qu.: 30.00
##  Max.   :40425    Max.   :105.00    Max.   :105.00    Max.   :118.00
```

```r
# boxplot of price by product category
boxplot(merged_frame$price ~ merged_frame$product_category_name,
        main = "Boxplot of Price by Product Category",
        xlab = "Product Category",
        ylab = "Price",
        las = 2, # Rotate x-axis labels for better readability
        cex.axis = 0.7) # Adjust x-axis label size
```



Histogram of price to see the distribution of prices in the dataset

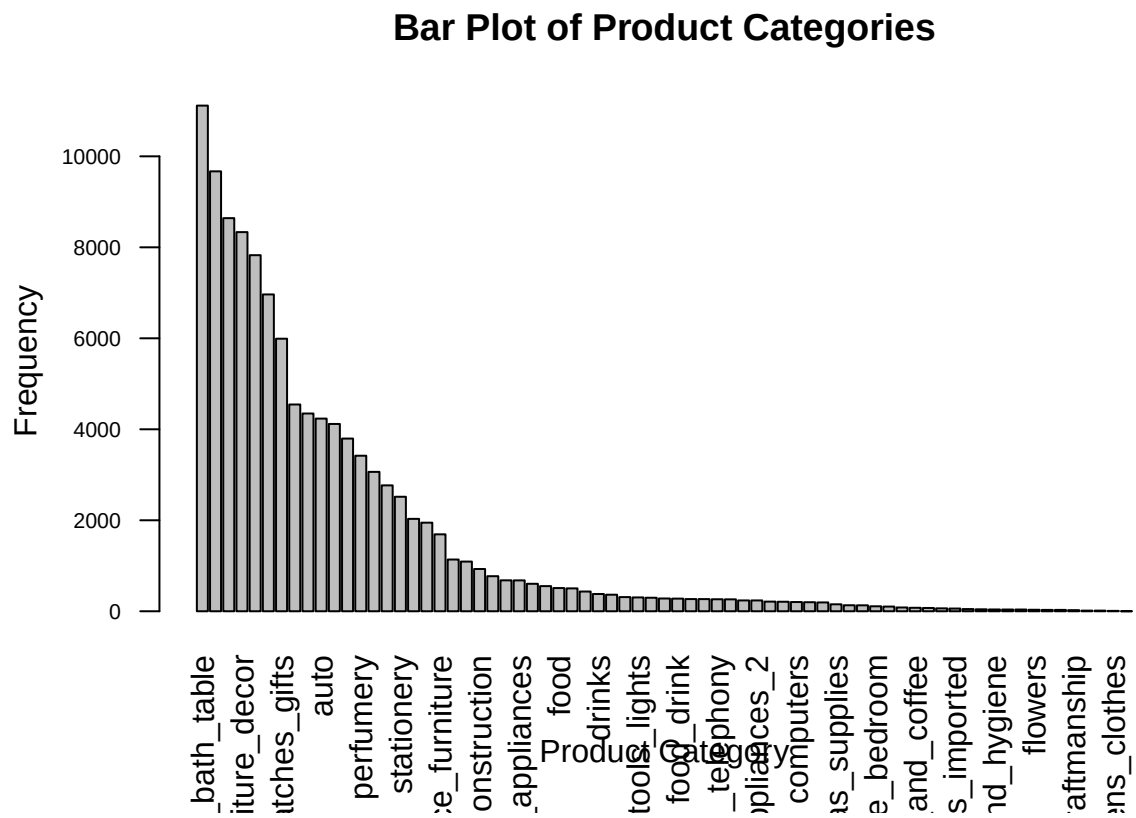
```
# Histogram of price
hist(merged_frame$price,
      main = "Histogram of Price",
      xlab = "Price",
      ylab = "Frequency",
      col = "lightblue",
      border = "black",
      breaks = 150)
```



Looking at how often each product category appears in the dataset

```
cbind(table(merged_frame$product_category_name))
```

```
##                                          [,1]
## agro_industry_and_commerce                212
## air_conditioning                          297
## art                                       209
## arts_and_craftmanship                      24
## audio                                     364
## auto                                     4235
## baby                                     3064
## bed_bath_table                          11115
## books_general_interest                    553
## books_imported                             60
## books_technical                           267
## cds_dvds_musicals                          14
## christmas_supplies                        153
## cine_photo                                 72
## computers                                 203
## computers_accessories                    7827
## consoles_games                           1137
## construction_tools_construction           929
```

```
## construction_tools_lights                 304
## construction_tools_safety                 194
## cool_stuff                               3796
## costruction_tools_garden                  238
## costruction_tools_tools                   103
## diapers_and_hygiene                        39
## drinks                                    379
## dvds_blu_ray                               64
## electronics                              2767
## fashio_female_clothing                     48
## fashion_bags_accessories                 2031
## fashion_childrens_clothes                   8
## fashion_male_clothing                     132
## fashion_shoes                             262
## fashion_sport                              30
## fashion_underwear_beach                   131
## fixed_telephony                           264
## flowers                                    33
## food                                      510
## food_drink                                278
## furniture_bedroom                         109
## furniture_decor                          8334
## furniture_living_room                     503
## furniture_mattress_and_upholstery          38
## garden_tools                             4347
## health_beauty                            9670
## home_appliances                           771
## home_appliances_2                         238
## home_comfort_2                             30
## home_confort                              434
## home_construction                         604
## housewares                               6964
## industry_commerce_and_business            268
## kitchen_dining_laundry_garden_furniture   281
## la_cuisine                                 14
## luggage_accessories                      1092
## market_place                              311
## music                                      38
## musical_instruments                       680
## office_furniture                         1691
## party_supplies                             43
## perfumery                                3419
## pet_shop                                 1947
## security_and_services                       2
## signaling_and_security                    199
## small_appliances                          679
## small_appliances_home_oven_and_coffee      76
## sports_leisure                           8641
## stationery                               2517
## tablets_printing_image                     83
## telephony                                4545
## toys                                     4117
## watches_gifts                            5991
```

```r
barplot(sort(table(merged_frame$product_category_name), decreasing = TRUE, las = 2),
        main = "Bar Plot of Product Categories",
        xlab = "Product Category",
        ylab = "Frequency",
        las = 2, # Rotate x-axis labels for better readability
        cex.axis = 0.7) # Adjust x-axis label size
```